

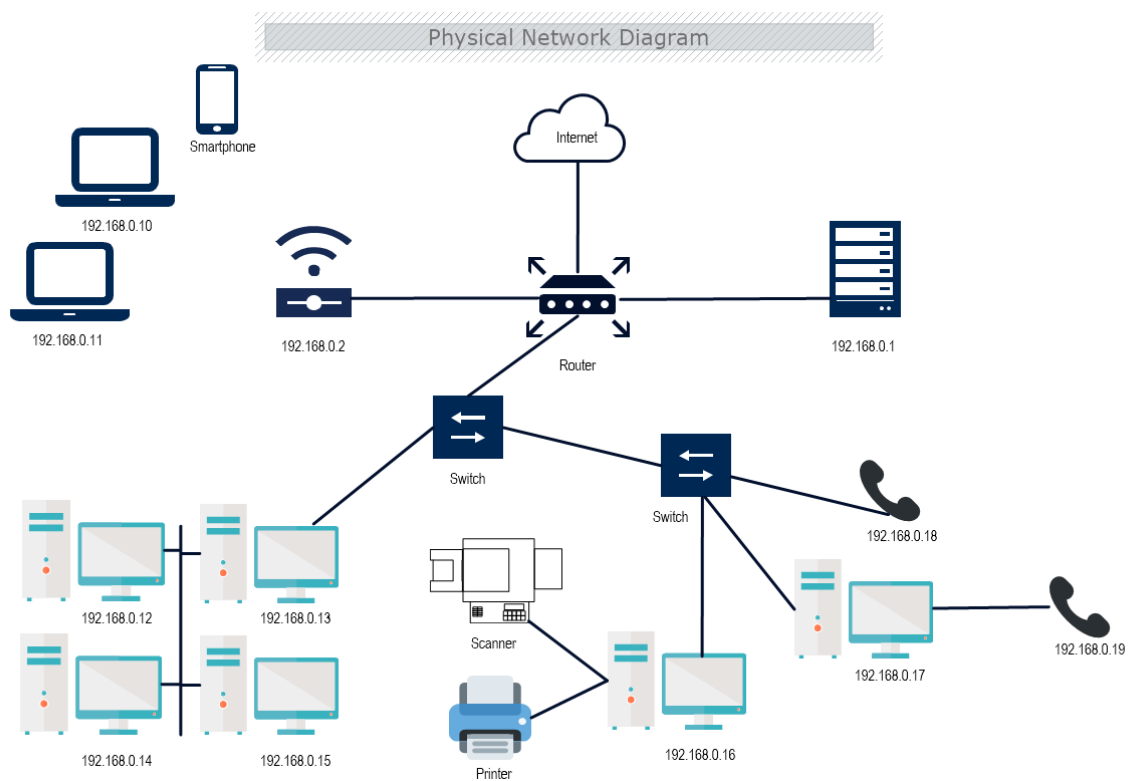
Physical Server

A **Physical Server** is the actual hardware machine.

It contains:

- CPU (Processor)
- RAM (Memory)
- Hard Disk/SSD (Storage)
- Network Card (NIC)

Without virtualization, one operating system runs directly on this server.



Advantages

- High performance
- No virtualization overhead

Disadvantages

- Hardware resources may be wasted.
- Only one OS runs at a time.
- Cost is higher.

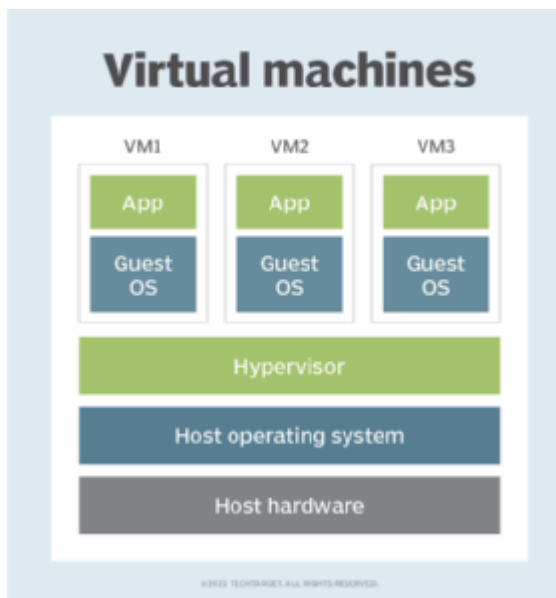
Virtual Server (Virtual Machine)

A **Virtual Server** (also called a **Virtual Machine - VM**) is a software-based computer created by the hypervisor.

Each VM behaves like an independent computer.

Each VM has:

- Own Operating System
- Own CPU allocation
- Own RAM allocation
- Own Storage
- Own Application



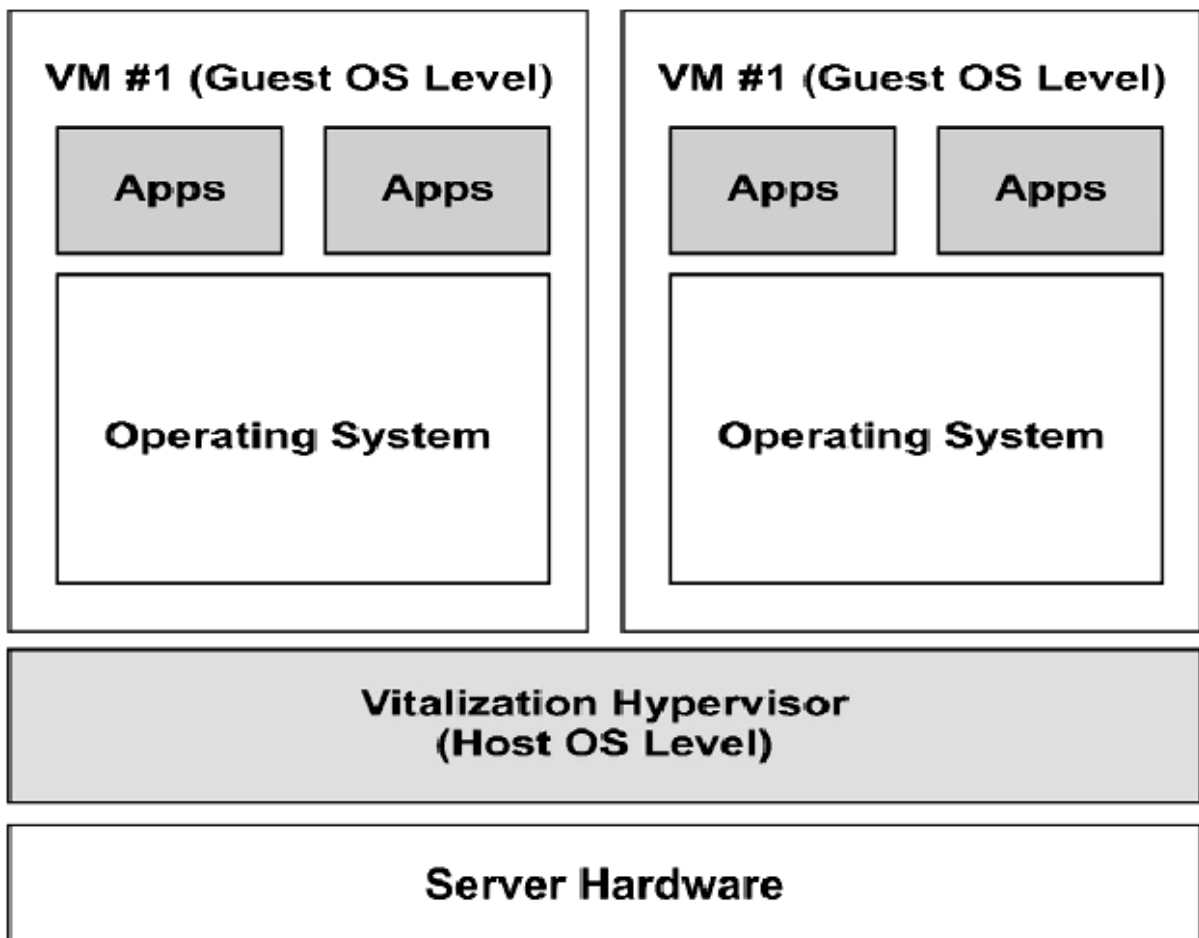
Hypervisor

A **Hypervisor** is software that creates and manages Virtual Machines (VMs).

It sits between the hardware and virtual machines.

Responsibilities

- Creates VMs
- Allocates CPU
- Allocates RAM
- Allocates Storage
- Shares hardware resources
- Starts and stops VMs



Types of Hypervisor

Type 1 (Bare Metal)

Runs directly on hardware.

Examples:

- VMware ESXi
- Microsoft Hyper-V
- Xen

Type 2 (Hosted)

Runs on top of an existing operating system.

Examples:

- Oracle VirtualBox
- VMware Workstation